Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/03/2005
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Project Title:	Radiation Transport Code Development for Space Radiation Shielding Applications		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Knoxville	**State:**	TN
Zip Code:	37996-2300	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	8	**No. of Master' Degrees:**	2
No. of Master's Candidates:	8	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pinsky, L. (University of Houston) Empl, A. (University of Houston) Ferrari, A. (CERN & INFN) Cerutti, F. (INFN) Garzelli, M.-V. (INFN) Ranft, J. (CERN & Univ. of Leipzig (Germany)) Gabriel, T. (Scientific Inv. & Develop.) Sihver, L. (Roanoke College/Chalmers University) Adams, J. (NASA MSFC) Watts, J. (NASA MSFC) Wilson, T. (NASA JSC) Moussa, H. M. (University of Tennessee) Qubain, E. (Roanoke College) Norbury, J. W. (Worcester Polytechnic Institute) Mayes, B (University of Houston)		
Grant/Contract No.:	NAG8-1901		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The task, to be carried out by a consortium of researchers from the University of Tennessee (lead institution), the University of Houston, Roanoke College, Worcester Polytechnic Institute and various government and national laboratories, involves the development of a standard set of radiation transport computer codes for NASA applications. This effort involves further improvements of the Monte Carlo codes HETC and FLUKA. Included in these improvements is the development of nuclear reaction databases necessary to extend these codes to heavy ion transport. Also planned are analyses of the remaining deficiencies in these codes and in the deterministic code HZETRN, and the determination of what is needed to develop a standard NASA radiation shielding code. Error and sensitivity analyses will be performed to determine errors induced in calculated particle spectra by uncertainties in fragmentation cross sections. The improved codes will be validated by comparisons with measured transport data as they become available. Methods of solving the Boltzmann equation for 3-D radiation transport will also be explored using the numerical method of lines approach. Finally, techniques to facilitate the geometric modeling of complex and detailed 3-D configurations such as spacecraft for use with these transport codes will be explored. To develop a standard space radiation transport code/set of codes for NASA applications. This goal will be accomplished by further development of the HZETRN, HETC and FLUKA radiation transport codes. The ability to simulate the radiation environment in space and the transport of these particles through spacecraft shielding materials is crucial to concerns such as crew health and electronic equipment performance. The need exists to understand both the actual measurement of crew radiation doses recorded on current missions as well as to explore the predicted doses for future missions and new spacecraft designs. These planning tasks call for significant flexibility to simulate a wide variety of conditions such as those likely to exist within spacecraft in interplanetary space, and on the surface of the Moon and Mars. In addition, many experiments need to simulate the effects that space radiation has on their instruments. These effects can either be direct as part of the actual measurements or as a background to it. Further, there can be operational concerns such as degradation of instruments from radiation damage or data interference as in the case of Single Event Upsets. The improvement in the accuracy of the software that NASA currently uses for these tasks that will result from this project will substantially increase the reliability of the projected results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Extensive testing, benchmarking and validation of the HETC-HEDS radiation transport computer code is being carried out by comparing predictions obtained from the code with measured particle cross sections and spectra obtained by the companion NASA Measurements Consortium, and by comparisons with published data from the scientific and engineering literature. In general the agreement between predictions and measurements for charged particle yields are good. Many, but not all of the predicted secondary particle yields agree to within 25% with the measured values. Comparisons of predicted neutron spectra to measured spectra from interactions of GCR-like ions with thick carbon targets showed significant disagreement between the code predictions and published measurements. Work is underway to determine the sources of these disagreements and to modify the code and/or event generator to improve the agreement. Comparisons of predicted charged particle yields from HETC-HEDS, for Fe beams at GCR energies interacting with thick polyethylene targets, were made with predictions obtained from the Japanese heavy ion transport code PHITS. The agreement between the predictions obtained from these two codes was good. 2. The HETC-HEDS code was modified to enable it to be used on personal computers utilizing Linux operating systems. 3. The HETC-HEDS and FLUKA codes were used as part of a suite of radiation transport codes in the design and characterization of a linear energy transfer (LET) spectrometer under development by a consortium involving Boston U., MIT, University of Tennessee, and Aerospace Corp. for use on the Lunar Reconnaissance Orbiter (LRO) spacecraft, which will be launched in late 2007/early 2008. During these analyses it was noted that the HETC-HEDS computational speed slowed significantly when applied to high-energy heavy ion transport, such as Fe at energies above 600 MeV/nucleon. The cause of this slowdown in computational speed is under investigation, but may be related to the methods used in the HZE particle event generator to conserve energy and momentum of the secondary particles produced in nuclear collisions of heavier nuclei, such as iron, at high energies. 4. Work on an improved, quantum mechanical nuclear fragmentation model for use in the HETC-HEDS HZE particle event generator is in progress. This effort uses the OPTFRAG knockout-ablation collision model, developed by the PI (Townsend) at NASA Langley some years ago. The intent is to replace the semiempirical NUCFRG2 model currently used in the event generator with the OPTFRAG model. Extensive comparisons of OPTFRAG model predictions of fragmentation cross sections are being made with the cross section data measured by the companion NASA measurements Consortium, and with available measured cross sections published in the open scientific and engineering literature. 5. A new institution and co-investigator (Dr. John Norbury, Worcester Polytechnic Institute) have been added to the Transport Code Development Consortium, with formal NASA approval, for the purpose of developing a suitable model of pion production from nucleus-nucleus collisions for inclusion in the HETC-HEDS event generator. 6. Work on the Numerical Method of Lines approach to solving the 3D Boltzmann transport equation at Roanoke College has been terminated. The method works well for one-dimensional transport, but does not work well for two-dimensional transport. Based upon the failure of the method in two dimensions, the decision was made to abandon any effort to extend it to three dimensions. Instead, approximate methods using spectral expansion techniques at intermediate energies and asymptotic expansions at high energies are being developed. 7. In January 2005, The UH Group sponsored and hosted the first FLUKA Course in the United States and the week-long program was attended by ~40 participants with 5 of the FLUKA principals as instructors, including Dr. Anton Empl, who is a member of the Houston group. A number of other members of the NASA Space Radiation Community, including participants from the Consortia themselves were present, along with participants from a number of US National Labs and several NASA Centers. 8.The UH Group has long advocated acquiring data in the energy range above 1 GeV/A for several reasons. Not only is the primary Cosmic Ray flux still significant through these higher energies, but there is a wide disagreement between existing nucleus-nucleus interaction models in this regime. As an outgrowth of discussions at last year's Space Radiation

Investigator's Workshop and at the COSPAR meeting in Paris a little over a year ago, the decision was taken to pursue measurements at the AGS this past summer before the AGS floor itself was to become unavailable for several years. To that end, and because of the lack of time and funding to develop additional detectors, The UH group agreed to employ existing detectors and expertise to enhance the date acquired during those runs. Additional funding was requested to cover the expenses, and in concert with the LBL Measurement Consortium Group, additional stipend money for a graduate student to work on these measurements as his Ph.D. thesis has been requested. In order to prepare for the AGS run, and to coordinate the setup and data acquisition details with the LBL Group, we also participated in their run at the HIMAC at NIRS in Chiba, Japan this past February. Professor Billy Mayes of the Physics Department at UH has joined our effort for these measurements.

9. Dr. Anton Empl working with a UH Undergraduate, Mr. Matthew Lebourgeois, have developed and released via the project website a ROOT-based GUI analysis tool to allow visualization of color-coded fluence plots superimposed on line drawings of the input geometry. Work is in progress to expand that capability to allow the plotting of all standard FLUKA scoring capabilities via the same ROOT-based GUI interface. Significant progress has also been made on the coding of an XML-based GUI front-end tool to enable the user to start and control FLUKA jobs completely from a GUI environment without the need for command-line interactions.

10.This past July, the long awaited release of the FLUKA source code under a CERN-INFN license agreement occurred, and along with that several improvements to the core FLUKA code were made, the most visible of which from the user's standpoint is the elimination of the need to make an "EGS-like" pre-run to setup the so-called "pemf" file necessary for the implementation of electromagnetic interactions. Now, that initialization activity is seamlessly embedded within the core FLUKA code.

11. Considerable work was done to implement and upgrade the treatment of electromagnetic dissociation in Nucleus-Nucleus interactions in FLUKA by Dr. Georgi Smirnov, making use of large volumes of data that are now accessible from Russian sources. The neutron cross section transport within FLUKA has been upgraded with the latest tables available from ENEA by Dr. GianCarlo Panini. This includes the use of these newly available inputs to both the Nucleus-Nucleus models and to the individual nucleon interactions. Mr. Neal Zapp has continued to develop his Hamiltonian Molecular Dynamics-based Nucleus-Nucleus Interaction code, and now has a working version that includes only the internal elastic scattering cross sections for the constituents. He is now in the process of exploring the introduction of the inelastic cross sections for the constituent particles. The FLUKA PI (Pinsky) has been working on the refinement of software tools to compare the direct outputs of the event generator models and to identify flaws in the outputs of the current versions of these codes that are employed within FLUKA. In addition, with the help of Mr. Kerry Lee, another UH Ph.D. thesis graduate student, models of Solar Modulation of the GCR flux are being explored. Likewise, Dr. Victor Andersen of the UH group has been involved in modeling SPEs for inclusion in the standard FLUKA input package.

12.Finally, efforts have been underway to not only improve the tools for the input of geometry information into FLUKA, but to develop a broad tool for general use in the conversion of CAD files into Monte Carlo Geometry input files. The Virtual Monte Carlo (VMC) project at CERN is progressing well and there is currently a version of FLUKA being tested in that new environment. The Generic ROOT-based geometry that is included in this package would be a strong candidate for the common Monte Carlo geometry to be the target of such a conversion software code.

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